Skills Summary

Perimeter and Area of Coordinate Polygons

Use this reference while completing your worksheet or when studying.

Distance (one side length): $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint: $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ **Slope:** $m = \frac{y_2-y_1}{x_2-x_1}$
Shoelace area: list the vertices in order around the polygon and take half the absolute value of $\sum(x_i y_{i+1} - x_{i+1} y_i)$. It works for any polygon, however slanted.

Which one, and why: distance builds a perimeter, shoelace gives an area directly, and slope or midpoint prove a claim about the shape rather than measure it.

1. Perimeter from coordinates

A side parallel to an axis has length equal to the difference of the two coordinates that change — no formula needed. Use the distance formula only on the slanted sides, then add every side once.

Two slanted sides of equal length appear as equal radicals, so a parallelogram or rhombus halves the work.

Example: Find the perimeter of the triangle with vertices $(1, 1)$, $(7, 1)$ and $(7, 9)$.
Step 1. Two sides are axis-parallel, so only the slanted side needs the distance formula — that is the one measurement the coordinates do not hand you.
Step 2. Horizontal side $7-1 = 6$; vertical side $9-1 = 8$; slanted side $\sqrt{6^2 + 8^2} = \sqrt{100} = \mathbf{10}$.
Perimeter $= 6 + 8 + 10 = \mathbf{24}$.

Try it: Find the slanted side and the perimeter of the triangle with vertices $(0, 0)$, $(9, 0)$ and $(9, 12)$.

check: slanted side **15**, perimeter **36**

2. Area from coordinates

Shoelace never asks for a side length, so it is the safe default when no side is horizontal or vertical. Write the vertices in boundary order, multiply down-right, subtract down-left, halve the absolute value.

Sanity check: the polygon is always smaller than the smallest rectangle that surrounds it, and a triangle is at most half of it.

Example: Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$ and $(2, 5)$.
Step 1. The apex sits over the middle of the base, so there is no clean rectangle to subtract from — take the vertices in order and use shoelace.
Step 2. $\frac{1}{2}\,|\,0(0-5) + 6(5-0) + 2(0-0)\,|$.
The area is **15** square units.

Try it: Find the area of the quadrilateral with vertices $(0,0)$, $(6,0)$, $(7,4)$ and $(1,4)$.

check: **24**

3. Justify the shape, then measure it

Slopes whose product is -1 mean perpendicular sides, so a right angle is established rather than assumed — and a right angle turns two sides into a base and a height.

Diagonals with the same midpoint bisect each other. Neither fact is a measurement; both are what makes a measuring shortcut legal.

Example: Triangle $(0,0)$, $(4,2)$, $(2,6)$. Is the corner at $(4,2)$ a right angle, and what is the area?
Step 1. Two sides meet at that corner, so their slopes decide the angle before any area work begins.
Step 2. $\mathbf{\frac{1}{2}}$ and $\mathbf{-2}$ are the two slopes, and their product is -1, so the sides are perpendicular; shoelace on the three vertices then gives **10** square units.

Try it: Triangle $(0,0)$, $(6,3)$, $(3,9)$. Find the slope of the side from $(0,0)$ to $(6,3)$, and the area of the triangle.

check: slope $\mathbf{\frac{1}{2}}$, area **22.5**

(!) Watch out: shoelace only works if the vertices are listed in order around the boundary. Jumping across a diagonal folds the polygon and returns a smaller number, so read your vertex list off the plot, not off the problem.